(a) Monthly Row-Contribution Fingerprint & Evidence Graph G

`fp = SUM(hash(K, ym, e))`

`store_sales` (grain K: ticket+item)

K (tkt,item)	ym (τ)	store	paid (e)
101, A	02-09	S1	50
102, B	02-09	NULL	30
103, A	02-10	S2	20

Three definitions of "revenue" (`SUM(paid)`)

- X: `FROM store_sales` (no dim join) — **keeps all 3 rows**
- Y: `JOIN store` (inner join on store) — **drops `102B` (`store=NULL`)**
- Z: `JOIN item` (inner join on item) — **every row has item (0 lost)**

Def	2002-09 Fingerprint	2002-10 Fingerprint	Relation in Graph G
X	$h(\texttt{101A}, 09, 50) + h(\texttt{102B}, 09, 30)$	$h(\texttt{103A}, 10, 20)$	**Base definition**
Z	$h(\texttt{101A}, 09, 50) + h(\texttt{102B}, 09, 30)$	$h(\texttt{103A}, 10, 20)$	**All months = $\longrightarrow$ X $\equiv_P$ Z (R2)**
Y	$h(\texttt{101A}, 09, 50)$ **(missing `102B`!)**	$h(\texttt{103A}, 10, 20)$	**09 $\neq$, 10 = $\longrightarrow$ $\delta = \{\mathcal{J}_{\text{store}}\}$**

✓ **Order-independent & incremental on ΔT (93 defs / 2.88M rows: 3.9 s; +6k rows: 0.24 s)**

3 (b) Online Constraint-Explaining Admission (No Upfront SQL Generation)

grounded Φ | **param p** | **derivation d** | **contrast δ** | **residue Res**

Case 1: Grounded hit or certified equivalent (X $\equiv_P$ Z)

"`highest month` [d] `of turnover` [Φ] `in 2002` [p]"

✓ Res $= \emptyset$, P **holds $\longrightarrow$ Bind (p, d) & reuse X directly**

Case 2: Unexplained modifier $\longrightarrow$ Partial hit

"`revenue` [Φ] `net of returns` [Res] `in Sep.` [p]"

✗ Res $= \{\text{net of returns}\} \neq \emptyset \longrightarrow$ **Partial hit (X + complete anti-join)**

Case 3: Underspecified request ($S = \{X, Y\}$) $\longrightarrow$ Period fingerprint check

Ask `2002-10`: fp(X, 10) = fp(Y, 10) | Ask `2002-09`: fp(X, 09) $\neq$ fp(Y, 09)

✓ `2002-10` **agrees $\longrightarrow$ return `20` without asking;** `2002-09` **differs $\longrightarrow$ ask 1 question on δ**

Case 4: Tentative similarity match $\longrightarrow$ 3-way audited risk bound

Unverified phrasing in stratum g: gate by confidence sequence $U_g(t) \leq \alpha$

✓ **If $U_g(t) \leq \alpha$: serve; else audit & label (equiv / distinct / unprovable)**